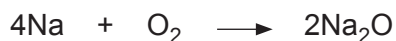


6. The balanced symbol equation for the reaction between sodium and oxygen is given below.



- (a) Calculate the mass of sodium oxide that should be produced when 0.736 g of sodium is burned in air. [3]

mass = g

- (b) When the reaction was carried out using 0.736 g of sodium, it was found that the mass of sodium oxide formed was 0.338 g.

- (i) Use your answer to part (a) to calculate the percentage yield of the reaction. [1]

percentage yield = %

- (ii) If no product was lost, suggest **one** possible explanation for the percentage yield not being 100%. [1]

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Examiner
only

6. Explain the trends in reactivity of the elements in Groups 1 and 7 of the Periodic Table. [6 QER]

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Examiner
only

6. Explain how the processes of boiling and adding washing soda remove hardness from water. [6 QER]
Include equations in your answer.

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Examiner
only

6. Levels of oxygen and carbon dioxide in the atmosphere are maintained at approximately constant values.

(a) Identify the natural processes that help maintain the balance of oxygen and carbon dioxide levels in the atmosphere. Describe briefly how this is achieved. [3]

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6. Fluoride ions are added to drinking water in some areas of the UK.

The table below shows the effect of different levels of fluoride in drinking water on the number of decayed, missing and filled teeth (DMFT) and the percentage of people suffering from fluorosis.

Concentration of fluoride (mg/dm ³)	Mean DMFT	Percentage of people affected by fluorosis (%)
0.3	7.0	4
0.6	4.5	6
0.9	3.0	15
1.2	2.5	35
1.5	2.3	40
1.8	2.0	45
2.1	2.1	60
2.4	2.1	68
2.7	2.1	75

- (a) The recommended concentration of fluoride ions to be added to drinking water in the UK is 1.0 mg/dm³.

Use the information in the table to explain why this is the recommended level in terms of DMFT and fluorosis. [4]

DMFT

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Fluorosis

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- (b) Chlorine is added to all drinking water in the UK, whilst fluoride is only added in some areas.

Explain why some people are opposed to adding fluoride but no-one is opposed to adding chlorine to drinking water. [2]

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- (c) Fluorine reacts with titanium at very high temperatures, forming titanium fluoride.

In an experiment 3.8 g of fluorine reacted with 2.4 g of titanium.

Calculate the simplest formula for titanium fluoride. You **must** show your working. [3]

$$A_r(\text{Ti}) = 48$$

$$A_r(\text{F}) = 19$$

Simplest formula

